

PHY201 Practice Exam on chapter(s): 11, 12

Name: \_\_\_\_\_

These questions assess your understanding of the material from Chapters 11, 12 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units.

1. The aorta pumps blood from the heart to the rest of the body. Calculate the average speed of the blood in Grace's aorta if the flow rate of her blood is 5.6 L/min. Grace's aorta has a radius of 14 mm.

ANSWER: \_\_\_\_\_

2. Suppose that Mark floats in freshwater with 96.8% of his volume submerged when his lungs are full of air. His mass is 58 kg. What is his average density?

ANSWER: \_\_\_\_\_

3. The aorta pumps blood from the heart to the rest of the body. The flow rate of blood through Sarah's aorta is 17.2 L/min and Sarah's aorta has a radius of 10 mm. Recall that blood flows through smaller blood vessels known as capillaries, which have an average diameter of 9.0  $\mu\text{m}$ . The speed of blood in the capillaries is about 0.32 mm/s. Calculate the number of capillaries in Sarah's blood circulatory system.

ANSWER: \_\_\_\_\_

4. Patricia attaches a nozzle of diameter 5.5 mm to a garden hose of diameter 1.5 cm. If the hose fills up a 1-L bucket in 11.3 seconds, calculate the pressure in the hose assuming level, frictionless flow.

ANSWER: \_\_\_\_\_

5. Isabella puts a coin on a scale and finds its mass is 6.39 g. When the coin and scale are completely submerged in water, the scale reports the coins mass as 2.42 g. What is the density of the coin?

ANSWER: \_\_\_\_\_

6. William attaches a nozzle of diameter 4.8 mm to a garden hose of diameter 2.9 cm. If the hose fills up a 2-L bucket in 14.6 seconds, what is the speed of the water that leaves the nozzle?

ANSWER: \_\_\_\_\_

7. While Allison is fishing, she begins to reel in a reel big fish. Her "FishFinder 3000" tells her that this particular fish has a volume of 6400  $\text{cm}^3$ . After Allison catches the fish and weighs it, she finds it has a mass of 7.18 kg. What was the minimum tension force in the fishing line if the fish was pulled vertically upwards through saltwater, which has a density of 1025  $\text{kg/m}^3$ ?

ANSWER: \_\_\_\_\_

8. Rachel is an astronaut working outside the International Space Station (ISS) where the atmospheric pressure is essentially zero. The pressure gauge on the astronaut's air tank reads  $5.6 \times 10^6$  Pa. What force does the air inside the tank exert on the flat end of the cylindrical tank, a disk that is 0.1 m in diameter?

ANSWER: \_\_\_\_\_

9. Calculate the depth below the surface of water at which the pressure due to the weight of the water equals 0.35 atm.

ANSWER: \_\_\_\_\_

10. A reservoir has a surface area of 37  $\text{km}^2$  and an average depth of 46 m. What mass of water is held behind the dam?

ANSWER: \_\_\_\_\_

11. Jessica turns a hose faucet on and after 6.6 seconds, sees water emerge from her 10-m-long garden hose of diameter 1.7 cm. Jessica then turns it off momentarily to attach a nozzle of diameter 7.3 mm to the end, then turns the water back on. She then uses the hose to fill up a 2-L bucket. What is the speed of the water that leaves the nozzle?  
ANSWER: \_\_\_\_\_
12. Jennifer's doctor performs a special test and tells her that a 2.8-N force can rupture her eardrum. If her eardrum has an area of  $0.89 \text{ cm}^2$ , at what depth in freshwater would her eardrum rupture, assuming the gauge pressure in the middle ear is zero and she does nothing to alleviate the pressure?  
ANSWER: \_\_\_\_\_
13. While Leah is fishing, she begins to reel in a reel big fish. Her "FishFinder 3000" tells her that this particular fish has a volume of  $9000 \text{ cm}^3$ . As she's reeling the fish in, she notes that her reel begins at rest and revolves 6 times in 8 seconds. The 480-g reel has a radius of 3 cm and a friction torque of 20 N cm. How much force did Leah apply to the reel's crank handle to reel the fish in if the handle is a distance of 5.4 cm from the center of the reel and the fish was pulled vertically upwards through freshwater, which has a density of  $1000 \text{ kg/m}^3$ ? Later measurements of the fish reveals it had a mass of 9.94 kg.  
ANSWER: \_\_\_\_\_
14. Shiv and Linda are on a slowly sinking ship, but they have hope: Shiv managed to find an oaken door that is 230-cm tall, 108-cm wide, and 7-cm deep. Perhaps they could both fit on the door if he throws it into the sea. What is the maximum mass the door will support? Linda recently took a physics class and remembers the density of oak is about  $890 \text{ kg/m}^3$  and the density of seawater is  $1025 \text{ kg/m}^3$ .  
ANSWER: \_\_\_\_\_
15. Isabella creates a soap bubble and measures its diameter to be 230  $\mu\text{m}$ . Calculate the absolute pressure inside of the bubble, given that the surface tension of typical soapy water is 0.0370 N/m. Report your answer in mmHg, recalling that 133.32 Pa is 1 mmHg.  
ANSWER: \_\_\_\_\_
16. William attaches a nozzle of diameter 4.8 mm to a garden hose of diameter 2.9 cm. If the hose fills up a 2-L bucket in 14.6 seconds, what is the speed of the water that leaves the nozzle?  
ANSWER: \_\_\_\_\_
17. Intravenous (IV) infusions are usually made with the help of the gravitational force. Assuming that the density of the fluid being administered is 1.00 g/mL, at what height should the IV bag be placed above the entry point so that the fluid only just enters the vein if the blood pressure in the vein is 12.3 mmHg above atmospheric pressure? Recall that 1.0 mmHg = 133.32 Pa.  
ANSWER: \_\_\_\_\_
18. A particular dam is 760-m wide and contains water that is 134-m deep. What is the average pressure on the dam due to the water?  
ANSWER: \_\_\_\_\_
19. A particular hydraulic system contains two fluid-filled cylinders which are connected by a tube. Cylinder 1 has a diameter of 2.2 cm and is capped with piston  $P_1$ . Cylinder 2 has a diameter of 6.9 cm and is capped with piston  $P_2$ . A force  $F_1$ , equal to 120 N, is exerted on  $P_1$ . What is the magnitude of the resulting force exerted by  $P_2$ ?  
ANSWER: \_\_\_\_\_
20. Calculate the maximum buoyant force that water could exert on a ship with a mass of  $2.6 \times 10^8 \text{ kg}$  if the ship displaces  $8.7 \times 10^6 \text{ m}^3$  of water.  
ANSWER: \_\_\_\_\_

21. While Ryan is fishing, he begins to reel in a reel big fish. His "FishFinder 3000" tells him that this particular fish has a volume of  $8800 \text{ cm}^3$ . As he's reeling the fish in, he notes that his reel begins at rest and revolves 79 times in 3 seconds. The 420-g reel has a radius of 5 cm and a friction torque of 26 N cm. How much force did Ryan apply to the reel's crank handle to reel the fish in if the handle is a distance of 6.5 cm from the center of the reel and the fish was pulled through saltwater, which has a density of  $1025 \text{ kg/m}^3$ , at an angle of  $49^\circ$  from the vertical? Later measurements of the fish reveals it had a mass of 9.6 kg.

ANSWER: \_\_\_\_\_